STARSYN AI · ACADEMY

# Complete AI *Curriculum*

Sixteen modules from first principles to a deployed system, taught to eight people at a time so every build gets real attention.

| | | | |
|---|---|---|---|
| **16** | **480** | **8** | **1** |
| MODULES | HOURS | SEATS | CAPSTONE |

StarSyn AI · Bangalore · Coimbatore · Nilgiris

Edition 2026.1

starsynai.com

# Four levels, in this order, for a reason

Nobody arrives able to build; everybody arrives able to understand. Skipping a rung is the most common reason people stall.

1 **UNDERSTAND AI**
What it is, what it is not, and why it fails
AI fundamentals · ML/DL/GenAI · LLMs · terminology · how the tools actually work

2 **USE AI**
Daily work, done faster and to a higher standard
Prompt engineering · research · writing · slides · coding · data analysis · image and video · productivity

3 **BUILD WITH AI**
Stop being a user, start being a builder
Python · APIs · LLM APIs · embeddings · RAG · vector databases · agents · automation

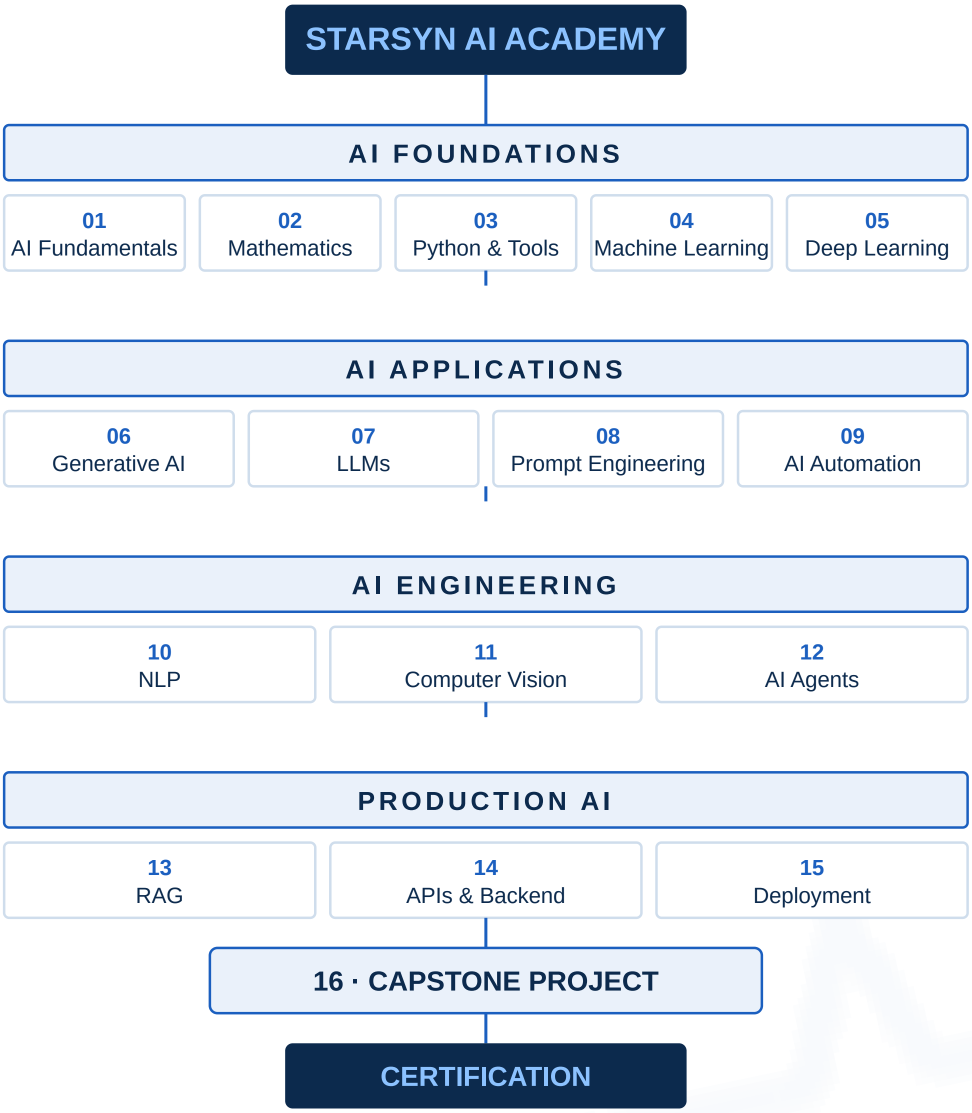
4 **SHIP PROJECTS**
Real systems, evaluated, deployed and defended
Chatbot · RAG application · research assistant · automation · SaaS prototype · capstone

| Who you are | Start | Finish | What you walk out holding |
|---|---|---|---|
| **Curious professional** | Level 1 | Level 2 | Prompt assets, a faster reporting routine, honest judgement about vendor claims |
| **Working specialist** EHS, quality, operations, admin | Level 1 | Level 3 | One recurring task automated on your own accounts, with an SOP a colleague can run |
| **Student or fresher** | Level 1 | Level 4 | A portfolio of built systems and a defended capstone, not a list of course names |
| **Developer** | Level 3 | Level 4 | A deployed, monitored, evaluated AI service with real users |
| **Company team** | Level 1–2 | Level 3 | Working internal tools, plus a written AI usage and data policy |

> **How every module is taught.** Learn the concept, see it fail on purpose, build it on your own data, evaluate it honestly, then hand it to somebody else.

# The complete structure

Four pillars, sixteen modules, one capstone. Each pillar depends on the one above it — applications make no sense without foundations, and production engineering makes no sense without both.




**Foundations first**

Modules 01–05 are non-negotiable beyond Level 2. They make the rest teachable rather than mimicked.

**Applications pay early**

Modules 06–09 produce visible results within days, which keeps a learner and their manager invested.

**Engineering separates**

Modules 10–12 are where a user becomes a builder. The ones who cross this line are employable.

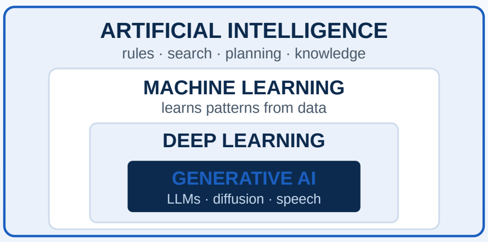
**Production is the point**

Modules 13–16 turn laptop work into something a business can depend on and pay for.

# AI Foundations

Modules 01–05. What these systems are, the mathematics and Python behind them, and the classical machine learning that still solves most real business problems.

| Module | What you learn & key algorithms | Tools | Flagship project |
|---|---|---|---|
| **01 · AI Fundamentals**<br>14 hours | The landscape · How learning happens · Vocabulary you will use daily · Limits and ethics<br>Rule-based systems · Search and planning · Decision trees | ChatGPT · Claude · Gemini | Map five tasks in your own work and classify each as automatable, assistable or unsuitable |
| **02 · Mathematics for AI**<br>22 hours | Linear algebra · Calculus · Probability and statistics · Optimisation<br>Gradient descent · SGD and mini-batch · Momentum | NumPy · SymPy · Matplotlib | Implement gradient descent from scratch in NumPy and plot the descent |
| **03 · Python & Engineering Tools**<br>26 hours | Python that matters · The data stack · Engineering habits · Working with services<br>NumPy vectorisation · Pandas groupby and merge · List and dict comprehension | Python 3.12 · Jupyter · VS Code | Clean and analyse a messy real-world CSV end to end |
| **04 · Machine Learning**<br>36 hours | Supervised — regression · Supervised — classification · Unsupervised · Doing it correctly<br>Linear · Logistic · Ridge · Lasso · kNN · Naive Bayes · SVM · Decision Tree · Random Forest | scikit-learn · XGBoost · LightGBM | Predictive maintenance model on machine sensor data |
| **05 · Deep Learning**<br>38 hours | Foundations · Training well · Architectures · Practice<br>MLP · Backpropagation · CNN · ResNet · U-Net | PyTorch · torchvision · Hugging Face | Build a neural network from scratch in NumPy, including backpropagation |



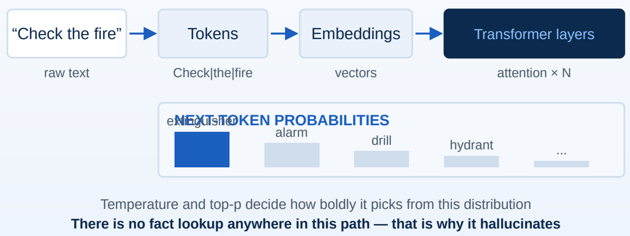

The nesting that clears up most confusion in the first hour

**What this pillar prevents.** Believing fluency means understanding, reaching for deep learning on 800 rows of tabular data, and reporting accuracy on an imbalanced safety dataset where predicting “no incident” scores 97%.

# AI Applications

Modules 06–09. Generation, large language models, prompting as an engineering discipline, and connecting all of it to the systems where work actually happens.

| Module | What you learn & key algorithms | Tools | Flagship project |
|---|---|---|---|
| **06 · Generative AI**<br>22 hours | Model families · Images · Video, audio, 3D · Responsible use<br>VAE · GAN · StyleGAN · DDPM · DDIM | Stable Diffusion · ComfyUI · Midjourney | Produce a complete branded visual set from a written brief |
| **07 · Large Language Models**<br>24 hours | Inside the model · How it was trained · Generation control · Judging quality<br>Byte-pair encoding · Self-attention · Multi-head attention | Anthropic API · OpenAI API · Gemini API | Build an evaluation set of 50 domain questions and score three models against it |
| **08 · Prompt Engineering**<br>24 hours | Core technique · Reasoning patterns · Engineering the prompt · Guarding it<br>Zero-shot · Few-shot · Chain of thought · Self-consistency | Claude · ChatGPT · Prompt registry | Build a prompt asset with a 20-input test set and a documented failure rate |
| **09 · AI Automation**<br>28 hours | Workflow thinking · The tools · AI in the loop · Running it for real<br>Event-driven triggers · Polling vs webhooks · Queues and batching | n8n · Make · Zapier | Automate one recurring task in your own job end to end, running on a schedule |



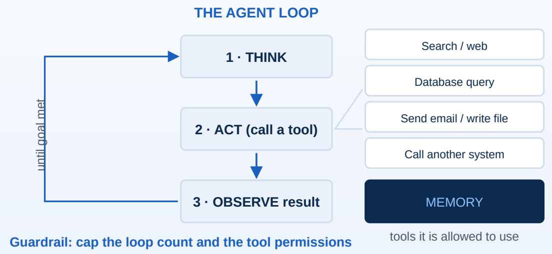

One request end to end — note that nothing in this path looks anything up

**What this pillar prevents.** Believing the model retrieved a fact, keeping prompts in a chat window where nobody else can run them, and pointing a first automation at a live production sheet.

# AI Engineering

Modules 10–12. Language, vision and autonomous agents — where a learner stops being a user of AI and becomes someone who builds with it.

| Module | What you learn & key algorithms | Tools | Flagship project |
|---|---|---|---|
| **10 · Natural Language Processing**<br>28 hours | Text preparation · Representation · Core tasks · Modern NLP<br>TF-IDF · word2vec · GloVe · fastText · Naive Bayes · SVM for text | spaCy · NLTK · scikit-learn | Build a text classifier on your own documents |
| **11 · Computer Vision**<br>32 hours | Image basics · Core tasks · Documents · Deployment reality<br>CNN · ResNet · EfficientNet · YOLO family · Faster R-CNN | OpenCV · YOLO (Ultralytics) · PyTorch | Build a PPE detection prototype on sample footage and report its false-positive rate |
| **12 · AI Agents**<br>34 hours | The loop · Memory · Structures · Control<br>ReAct · Plan-and-execute · Reflexion | LangGraph · Claude tool use · MCP | Build a tool-using research agent with a strict source-citation rule |




The agent loop, with limits on both the loop and the tools

**What this pillar prevents.** Sending every row to a paid API when a classifier would do it in milliseconds, demonstrating vision on clean stock images, and giving an agent unlimited tools and unlimited loops.

## WHAT A LEARNER CAN BUILD AFTER THIS PILLAR

- A classifier that sorts incoming reports by category and severity, running for a fraction of the cost of an API call per row
- An extractor that pulls names, dates and clause references out of unstructured documents
- A detection prototype that flags missing PPE on plant camera footage, with its false-positive rate measured
- A tool-using agent that gathers, checks and drafts a recurring pack, stopping at a human checkpoint

## WHY THIS IS THE LINE THAT SEPARATES PEOPLE

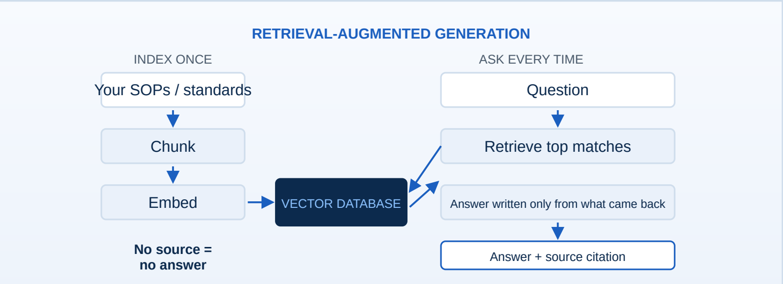
Everything before this pillar can be done by a capable person with a chat window. Nothing after it can. Modules 10 to 12 are where a learner stops asking a model for answers and starts designing systems that produce them: choosing a representation, training something on their own data, measuring whether it actually works, and deciding what an autonomous process is and is not permitted to do.

It is also where most self-taught learners stall, because the work now has to be evaluated. Every module here ends with a number — an accuracy, a false-positive rate, a cost per run — rather than a demonstration that felt impressive.

# Production AI

Modules 13–15. Grounding answers in your own documents, serving the result behind an API, and keeping it alive once real people depend on it.

| Module | What you learn & key algorithms | Tools | Flagship project |
|---|---|---|---|
| **13 · Retrieval-Augmented Generation** 34 hours | Indexing · Storage and search · Generation · Evaluation Chunking with overlap · Dense embeddings · Cosine / dot-product search | pgvector · Chroma · FAISS | Build a RAG system over your own document set with mandatory citations |
| **14 · APIs & Backend** 30 hours | Service basics · Working with model APIs · Data and state · Security REST · JSON Schema · JWT / API key auth | FastAPI · Pydantic · Postgres | Wrap a working model in a documented FastAPI service with auth |
| **15 · Deployment & Operations** 28 hours | Packaging · Shipping · Operating · Staying good Containerisation · CI/CD pipelines · Blue-green deploys | Docker · GitHub Actions · Render | Containerise and deploy one of your builds with a public endpoint |




Index once on the left, answer with sources on the right

**What this pillar prevents.** Skipping evaluation because the answers look right, putting an API key in front-end code, and declaring victory at launch while quality quietly drifts.

## WHAT A LEARNER CAN BUILD AFTER THIS PILLAR

- A question-answering system over the organisation's own standards and SOPs that cites the clause it answered from
- An authenticated API serving that system to a whole department, with streaming, caching and cost logging
- A deployment pipeline that ships on every push and rolls back cleanly when something breaks
- Monitoring that reports latency, cost, failures and answer quality, with alerts before a user complains

## WHY MOST AI PROJECTS DIE HERE

They do not die during the build. They die three months after launch, when the document set has changed, an upstream model has been deprecated, the monthly bill has quietly tripled, and nobody is measuring answer quality any more. The organisation concludes that AI did not work.

This pillar engineers against that outcome. Every system built here ships with an evaluation set, a cost figure, a named owner and a runbook, so whoever inherits it can keep it alive.

# Module 16 — the capstone is the certificate

One system, chosen by you, built end to end, evaluated honestly and defended live. Everything before it is preparation for this.

## SUGGESTED CAPSTONE TRACKS

- Document intelligence over a real archive
- Assistant grounded in real policy
- Automation of a recurring report
- Vision inspection or register digitising

## WHAT YOU SUBMIT

- Working system, running unattended
- Repository with README and SOP
- Evaluation against a test set

## ASSESSMENT ACROSS EVERY FORMAT

| Component | Weight |
|---|---|
| Module quizzes | 10% |
| Assignments and coding labs | 25% |
| Mid-programme practical | 15% |
| Final examination | 15% |
| Capstone project | 35% |

**The pass bar.** 70% overall, and a system that ran unattended on real data. Scoring well on paper with nothing running earns no certificate.

## HOW THE CAPSTONE RUNS

| Stage | What happens | Reviewed before you continue |
|---|---|---|
| **Weeks 1–2** Scope | Problem stated, process baselined, data confirmed, architecture agreed | Is the problem real, is the data available, can it be finished in time? Projects fail here more than anywhere else, so nothing is built yet. |
| **Weeks 3–6** Build | Built in weekly increments, each demonstrated to the cohort | Does it run end to end on real data, however roughly? Nothing running by week 4 means the scope gets cut, not extended. |
| **Week 7** Prove | Evaluation scored, failures documented, SOP written | Can you state how well it works with a number, and name what you would not trust it with? |
| **Week 8** Defend | Live demonstration to the cohort and an outside guest | It runs live, the figures hold up, and the design survives challenge. |

## WHAT THE CERTIFICATE STATES

This certifies that **[Name]** completed **[Programme]**, comprising [XX] hours of instruction and assessed practical work. Capstone built: **[System name]**. Measured result: **[before]** → **[after]**. Assessment score: **[XX]%**.

Certificate ID · Issued [date] · Instructor: Hariharan, Founder, StarSyn AI · Verify at starsynai.com/verify/[ID] · Issued by StarSyn AI, a private training provider; not a government-recognised or academic qualification.

# How the same curriculum is delivered

Every format is capped at eight participants. Above that the instructor cannot look over each person's shoulder.

| Format | Modules | Duration | Built for |
|---|---|---|---|
| **Applied AI Build Cohort** | 01, 07, 08, 09, 13 | 4 weeks · 8 seats | Professionals automating one real task from their own job |
| **Corporate workshop** | 01, 08, 09 | 1 day · 8 seats | A department that needs everyone capable by next week |
| **AI Career Track** | 01–16 complete | 6 months · 8 seats | Students and career changers building a portfolio |
| **LLM Engineering Track** | 03, 05, 07, 12, 13, 14, 15 | 10–12 weeks · 8 seats | Developers moving into AI engineering roles |

## WHERE THE 480 HOURS GO

AI Foundations · 01–05 · 136 hrs | AI Applications · 06–09 · 98 hrs | AI Engineering · 10–12 · 94 hrs | Production AI · 13–15 · 92 hrs | Capstone · 16 · 60 hrs | **Complete curriculum · 480 hrs**

**On sources.** Built against MIT OpenCourseWare, Stanford CS229, CS231n and CS224n, the Hugging Face courses, fast.ai, and the PyTorch and scikit-learn docs. These set depth and sequencing only; every slide, note and assessment here is written by StarSyn AI.

## HOW ENROLMENT WORKS

- A short call: what you do, what you would bring, whether this format fits
- Dates, tooling access and data-handling rules agreed in writing
- Payment before session one; full refund if you withdraw before session two

## WHAT THIS CERTIFICATE IS, AND IS NOT

A private professional certificate naming the system you built and the result you measured, verifiable online. It carries the weight of evidence, not of accreditation.

Not a degree, diploma or government-recognised qualification, and not represented as one.

### Next step

- Batch 01 opens with eight seats. Selection is by interview, not by payment.
- Every participant brings one real recurring task from their own job. No task, no seat.
- Enquiries: starsynai.com · Bangalore · Coimbatore · Nilgiris